Wu 2020 AML — analysis summary

RUN 1 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ=10 on sample (iLISI 9.37 of 23.63 ceiling climbs to θ=10, drops at θ=20). After removing 20 cells in 17 detached specks, all 96,881 cells form ONE connected component: samples intermix per-neighbourhood on a single continuous manifold. Judged on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

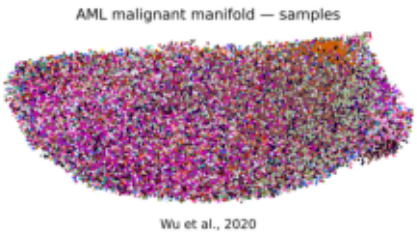
Single Gast/pSom-high → Crest-high developmental axis. Organized Gast↔pSom cross-repression (smoothed r=−0.68); Crest opposes Gast (r=−0.67), co-varies weakly with pSom (r=+0.11) and rises to the terminus (Crest-high decile at pseudotime 0.82; Spearman ρ=+0.43). No Gast/pSom co-expression hub and no opposing Gast arm — non-branching, not a bifurcation. Crest terminal, not on the Gast arm (no inversion flag).

Gate 3 — hub cross-repression → NA

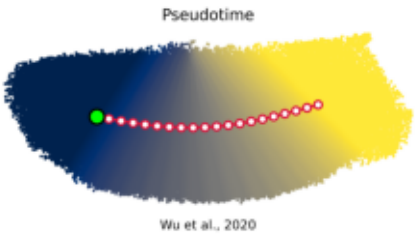
Branching-only. The manifold does not bifurcate — the diagnostic tree fit shows no Gast/pSom co-expression hub and no opposing Gast arm (both non-root tips are pSom-high / Crest-low), and the committed curve is fork-free. There is no hub population to test, so Gate 3 does not apply.

STEP FIGURES

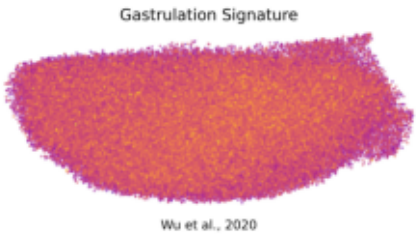
Manifold — samples



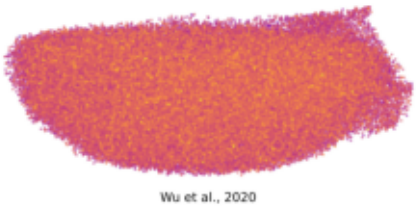
Pseudotime



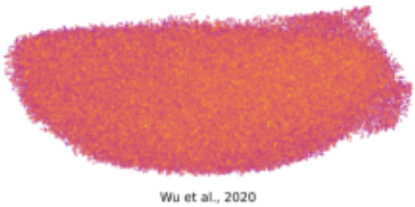
Gastrulation



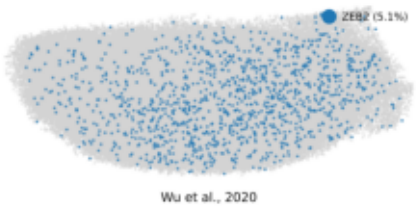
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute myeloid leukemia (Wu et al., 2020), Microwell-seq — 96,881 malignant cells (36 samples); single low-cycling body (cycling fraction 0.001).

Verdict.
The embryonic-program pattern reproduces in a NON-BRANCHING form: one continuous, sample-integrated AML manifold carrying a single Gastrulation/Somitogenesis-high → Neural-crest-high developmental axis with organized Gast↔pSom cross-repression (r=−0.68) and a terminal Crest gradient. No bifurcation, so Gate 3 is NA. Gate 1 & 2 GO; geometry Linear-pSom/Crest. EMT-TF ZEB2 (5.1%) is the sole detectable canonical factor.

Parameter choices

Min-cell threshold	100 (floor; no dominant cliff) → 36/38 samples, 96,901 cells (dropped P18, P13)	**Integration**	Harmony θ=10, n_pcs=20, k=30, spread=1.3 (iLISI peaks θ=10, falls θ=20); 443 overdispersed
Island removal	20 cells / 17 specks (UMAP radius graph 6× median NN) → 96,881	**Geometry**	Non-branching curve N=20, fold-free (SA 2.90); |corr(t, DC1)|=0.91
Root	curve tip at Gast/pSom-high, Crest-low pole; pt vs cycling ρ=−0.07	**Module means (raw)**	pSom +0.007 > Gast +0.002 > Crest −0.004 (Linear-pSom/Crest)
EMT-TF readout	ZEB2 (5.1%) — sole detectable canonical EMT-TF; broad panel below floor/absent		